ABSTRACT ALGEBRA 1 - HOMEWORK 7- M. VICARÍA

Dear student please submit the complete solution to each of the following problems in Gradescope in a PDF file. Please write down your name in each page and specify clearly the problem that you stated. For your study purposes, I included a classification of the problems and their difficulties:

- Warm-up: these are require that you understand some definition.
- *: they indicate some medium level of difficulty. You should spend longer in these problems.
- ** indicates that the problem is challenging, they will require more time than the other problems.

The complexity from ** and * is the level we would like you to master in this course. Questions highlighted with **OPTIONAL** are not necessary to get full credit for the homework, but I highly encourage you to solve them for your learning process.

In each of the solutions you SHOULD JUSTIFY COMPLETELY YOUR ANSWER with a proof or a counter-example. No credit is granted to solutions without justification.

1. Mandatory Problems

Problem 1^***:** (5 points) Consider $G = \mathrm{GL}_2(\mathbb{R})$ together with matrix multiplication. Let:

- $H_1 = \{A \in G \mid A \text{ diagonal}\}$,
- $H_2 = \{A \in G \mid \exists\lambda \in \mathbb{R}\ (A = \lambda \mathrm{I}_2)\}$, where I_2 is the identity 2×2 matrix,
- $\mathrm{SL}_2(\mathbb{R}) = \{A \in G \mid \det(A) = 1\}$.

(1) (1 point) Show that H_1 is not normal.
(2) (1 point) Show that H_2 is normal.
(3) (2 points) Show that $G/\mathrm{SL}_2(\mathbb{R}) \cong (\mathbb{R} \setminus \{0\}, \cdot\ )$ (hint: Isomorphism Theorem).
(4) (1 point) Show that $H_2 \cong G/\mathrm{SL}_2(\mathbb{R})$.

Problem 2^***:**(3points)Let $(G, \cdot_G, e_G)$ be a group.

(1) (1 point) Let H be a subgroup, show that the following statements are equivalent (this problem gives several ways to think about normal subgroups)
  (a) for all $g \in G$, $h \in H$, $ghg^{-1} \in H$,
  (b) for all $g \in G$, $gHg^{-1} = H$,
  (c) for all $g \in G$, $gH = Hg$ for all $g \in G$.

(2) (1point)In the following examples show that H is a normal subgroup of G, compute the factor quotient G/H and determine the order of G/H.
  - $G = \mathbb{Z}_4 \times \mathbb{Z}_2$ and $H =< (2,1) >$,
  - $G = (\mathbb{Z}_2 \times S_3)$ and $H =< (1, \rho_1) >$,
  - $G = \mathbb{Z}_3 \times \mathbb{Z}_5$ and $H = 0 \times \mathbb{Z}_5$.

(3) (1point)Let $(K, \star, e_K)$ be a group and $\phi : G \mapsto K$ a surjective homomorphism. Show that ϕ is an isomorphism if and if $Ker(\phi) = \{e_G\}$.

Problem 3**(2 points) For a group G, and a subset $X \subset G$, we define

$$\mathrm{UG}(G) := \{H \in \mathcal{P}(G) : H \text{ is a subgroup of } G\}$$

to be the set of all subgroups of G, and

$$\mathrm{UG}_X(G) := \{H \in \mathrm{UG}(G) : H \supseteq X\}$$

to be the set of all subgroups of G containing X. Let N be a normal subgroup of G, and consider the epimorphism $\pi\colon G \to G/N, g \mapsto gN$. Show that the map

$$\begin{aligned} \mathrm{UG}(G/N) &\to \mathrm{UG}_N(G) \\ \bar{H} &\mapsto \pi^{-1}(H) \end{aligned}$$

is a bijection.

Problem 4*: (5 points) Let $m \in \mathbb{N} \setminus \{0\}$. Consider the epimorphism $\pi \colon \mathbb{Z} \to \mathbb{Z}/m\mathbb{Z}$.

(1) (2 points) List all subgroups of $\mathbb{Z}$ containing $m\mathbb{Z}$.
(2) (2 points) Show that
$$\begin{aligned} \mathrm{UG}(\mathbb{Z}/m\mathbb{Z}) &\to \mathrm{UG}_{m\mathbb{Z}}(\mathbb{Z}) \\ \bar{H} &\mapsto \pi^{-1}(\bar{H}) \end{aligned}$$
is a bijection. (Hint: Find an inverse to this map.)
(3) (1 point) List all subgroups of $\mathbb{Z}/m\mathbb{Z}$.

2. Optional Problems

Problem 1* Let $(G, \cdot)$ be a group and let $H' \subseteq H$ be subgroups of G.

(1) (2 point) If $[G : H] = 2$ then show that H is normal in G.
(2) (2 points) If $[G : H] = 3$ and there exists an element $x \in G \setminus H$ such that $Hx = xH$, then show that H is normal in G.

[Problem 2*] Let $Y \subset X$ be sets, let $(G, *)$ be a group and let $(G^X, *)$ be the group of functions $X \to G$ where $*$ is defined pointwise. Let
$$N := \left\{ f \in G^X : f(y) = 1 \text{ for all } y \in Y \right\},$$
where 1 is the neutral element in $(G, *)$.

Show that $G^X/N \cong G^Y$.